Software Requirements Specification (SRS) for FLARE

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Revision History

Version	Date	Author	Description
0.1	2026-10-05	FLARE Team	Initial Draft
1.0	2026-10-05	FLARE Team	Final Release

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, interface, and operational requirements for FLARE - an enterprise-grade, offline-first application architecture, local data synchronization engine, and edge intelligence daemon.

FLARE is designed for environments with intermittent, unreliable, or untrusted network connectivity, including regional educational systems, mobile money micro-services, local enterprise nodes, and field-deployed edge devices.

1.2 Scope

FLARE provides:

- Offline-first data persistence with local SQLite (WAL mode) as the source of truth
- Local-first application runtime with optimistic UI via IndexedDB/Dexie.js
- Background daemon (`flared`) for orchestration, scheduling, and sync
- Natural Git commit engine for human-like local commit history
- Cryptographic node-locking (Ed25519), zero-trust authentication, and payload encryption (AES-256-GCM)
- Asynchronous sync with exponential backoff when connectivity is available
- Unix domain socket IPC with optional HTTPS bridge for local clients
- Cross-platform frontend (React + TypeScript + Vite + Tailwind CSS)
- Production-ready operations via systemd, encrypted backups, and observability

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
FLARE	Offline-First, Edge-Resilient Application Architecture
flared	FLARE daemon (background service)
IPC	Inter-Process Communication
WAL	Write-Ahead Logging (SQLite)
CRDT	Conflict-Free Replicated Data Type
Ed25519	Edwards-curve digital signature algorithm
AES-256-GCM	Advanced Encryption Standard with Galois/Counter Mode
mTLS	Mutual Transport Layer Security
SRS	Software Requirements Specification

1.4 References

- FLARE Technical Specification (Flare.md)
- SQLite WAL Documentation
- Ed25519 (RFC 8032)
- NIST AES-GCM
- Dexie.js Documentation
- FastAPI/Uvicorn Documentation

1.5 Overview

This document follows IEEE-style SRS structure covering system purpose, constraints, interfaces, detailed features, and quality attributes.

2. Overall Description

2.1 Product Perspective

FLARE is a self-contained, local-first system composed of multiple integrated components:

- **Frontend Layer**: React/Vite/TypeScript/Tailwind with Dexie.js for offline IndexedDB
- **Daemon Layer** (`flared`): Go-based background service managing state, licensing, sync, Git
- **Crypto Layer**: C++ shared library for Ed25519, AES-256-GCM, hardware fingerprinting
- **API Layer**: Django REST API for management, audit, and sync queue access
- **Bridge Layer**: FastAPI HTTPS bridge providing secure local access to IPC
- **Data Layer**: SQLite (WAL) as local authority; optional remote replication targets

2.2 Product Functions

- Deterministic local state management with offline authority
- Natural Git commit generation with time-based format rotation
- Node-locked license verification and hardware fingerprinting
- Offline sync queue with retry, backpressure, and exponential backoff
- Secure IPC over Unix domain socket (loopback-only bridge)
- Encrypted local storage awareness and backup/restore
- Audit logging with hash-chain integrity
- Systemd integration for production resilience

2.3 User Classes and Characteristics

Class	Characteristics
System Operator	Deploys, monitors, manages services via CLI/systemd
Developer	Extends/maintains codebase, runs local dev/test
End User	Interacts with FLARE UI in offline/online modes
Admin	Manages node config, audit logs via Django Admin

2.4 Operating Environment

- **Platform**: Linux (primary target; design supports cross-platform)
- **Runtime**: Go 1.20+, Python 3.10+, Node 18+, C++17
- **Services**: systemd (production), Unix domain sockets
- **Storage**: Local filesystem (~/.flare-data), SQLite WAL

- **Network**: Optional (loopback-only for local clients; outbound only on sync)

2.5 Design and Implementation Constraints

- Offline-first: core functionality must work without internet
- No hard dependency on external cloud during normal operation
- Use local cryptographic identity (Ed25519 keypair)
- Deterministic, reproducible builds
- Secure defaults: loopback-only, token-based auth for bridge
- Self-contained deployment with minimal external deps

2.6 Assumptions and Dependencies

- Local filesystem writable for data dir and logs
- systemd available for production service management
- OpenSSL available for bridge TLS cert generation
- Git installed if Natural Commit Engine used
- SQLite3 available for maintenance/backup

2.7 Apportioning of Requirements

Future releases may extend to P2P sync, multi-node clustering, and additional platform support.

3. External Interface Requirements

3.1 User Interfaces

- **Web UI (Frontend)**: Vite/React SPA at http://127.0.0.1:5173 (dev) or http://localhost:8080 (Docker)
- **Dashboard**: System status, daemon connectivity, offline queue metrics, Dexie item counts
- **About Page**: Product branding (FLARE) + Partner branding (BlueRhea Technologies) with theme-aware logos
- **Django Admin**: http://127.0.0.1:8000/admin/ for node config, audit logs, sync queue

3.2 Hardware Interfaces

- Local storage (filesystem)
- Network interface (optional, outbound for sync when online)

3.3 Software Interfaces

Interface	Protocol	Purpose
Unix Domain Socket	JSON line-based	flared IPC at /tmp/flared.sock
HTTPS Bridge	REST/JSON over TLS	Local loopback access to IPC (Bearer token)
Django REST API	HTTP/JSON	/api/health, /api/sync/queue, /api/sync/pull
Git CLI	subprocess	Natural commit operations
SQLite	WAL	Local persistence

3.4 Communications Interfaces

- Unix socket (AF_UNIX) for local IPC
- HTTPS (TLS 1.2+) for bridge on 127.0.0.1:8443
- HTTP for Django/API and Vite dev (loopback only)
- Optional WebSocket (WSS) for remote sync (per spec)

4. System Features

4.1 Offline-First Data Persistence

Description: Local SQLite database with WAL mode serves as immutable local authority.

- FR-1: All reads/writes must succeed offline
- FR-2: WAL mode enabled with synchronous=NORMAL
- FR-3: Foreign keys enforced
- FR-4: Automatic recovery on restart
- FR-5: Atomic snapshots via backup script

4.2 flared Daemon (Background Engine)

Description: Persistent background service orchestrating core logic.

- FR-6: Auto-start on boot via systemd
- FR-7: Graceful shutdown on SIGTERM/SIGINT
- FR-8: Health/status reporting over IPC
- FR-9: Background sync worker with periodic ticks
- FR-10: Robust restart on failure

4.3 Natural Git Commit Engine

Description: Generates human-like commit messages with time-based rotation.

- FR-11: Detect unstaged/staged changes via git status
- FR-12: Stage changes (git add -A)
- FR-13: Rotate formats every 2 hours: CASUAL*SHORT, CONVENTIONAL, TWO*LINE
- FR-14: Avoid automation/tool markers
- FR-15: Two-line commits reference last commit context
- FR-16: Local commits succeed offline

4.4 Offline Sync Queue

Description: Durable queue for pending sync operations.

- FR-17: Persist queue items to SQLite
- FR-18: States: PENDING, PROCESSING, COMPLETED, FAILED
- FR-19: Retry with exponential backoff + jitter
- FR-20: Max attempts configurable
- FR-21: IPC endpoints: enqueue, pull_queue
- FR-22: Idempotent processing semantics

4.5 Cryptography & Node-Locking

Description: Ed25519 + AES-256-GCM with hardware fingerprinting.

- FR-23: Ed25519 keypair generation
- FR-24: Message signing/verification
- FR-25: AES-256-GCM encrypt/decrypt
- FR-26: Hardware ID derivation
- FR-27: Node-locked license verification
- FR-28: Audit log integrity via hash chaining

4.6 Secure IPC & HTTPS Bridge

Description: Local-only secure access.

- FR-29: Unix socket for local processes
- FR-30: HTTPS bridge on 127.0.0.1 only
- FR-31: Bearer token authentication
- FR-32: CORS restricted to localhost origins
- FR-33: IP/Trusted-host allowlists
- FR-34: Rate limiting per IP
- FR-35: Security headers (HSTS, CSP, etc.)

- FR-36: Timeouts, body/line limits, concurrency caps
- FR-37: Cmd allowlist enforcement
- FR-38: /health, /ready, /metrics, /branding endpoints

4.7 Frontend (React + Dexie)

Description: Offline-capable UI with branding.

- FR-39: IndexedDB via Dexie.js
- FR-40: Optimistic UI updates
- FR-41: Theme-aware FLARE + BlueRhea logos
- FR-42: Dashboard and About views
- FR-43: IPC client with HTTP fallback
- FR-44: Responsive Tailwind UI

4.8 Django API & Admin

Description: Management interface.

- FR-45: Models: system*node*config, local*audit*logs, offline*sync*queue
- FR-46: REST endpoints for health and sync
- FR-47: Branded Django Admin
- FR-48: Static logo assets served

4.9 Operations & Backup

Description: Production readiness.

- FR-49: systemd service units (hardened)
- FR-50: WAL-safe backup script
- FR-51: TLS cert generation for bridge
- FR-52: Docker compose for containerized testing
- FR-53: Local multi-process runner
- FR-54: Makefile targets

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-1: Daemon startup < 1s
- NFR-2: IPC roundtrip < 10ms (Unix socket)
- NFR-3: UI initial load < 2s (local dev)
- NFR-4: Sync queue processing non-blocking
- NFR-5: Concurrent IPC requests up to configurable limit

5.2 Safety Requirements

- NFR-6: Atomic SQLite writes via transactions
- NFR-7: Graceful degradation when remote unavailable
- NFR-8: No data loss on crash (WAL recovery)

5.3 Security Requirements

- NFR-9: Bridge bound to 127.0.0.1 only
- NFR-10: Token-based auth for bridge IPC endpoints
- NFR-11: Security headers enabled
- NFR-12: Rate limiting enforced
- NFR-13: Input validation on IPC commands (allowlist)
- NFR-14: TLS for bridge
- NFR-15: No secrets in logs
- NFR-16: Local-first identity (no forced cloud auth)

5.4 Software Quality Attributes

- NFR-17: Modularity (decoupled layers)
- NFR-18: Testability (local testability)
- NFR-19: Maintainability (clear structure)
- NFR-20: Portability (POSIX/Linux primary)
- NFR-21: Reliability (auto-restart, offline-first)

5.5 Business Rules

- BR-1: Local data is source of truth
- BR-2: Remote is async replication target
- BR-3: Natural commit history preserved
- BR-4: Branding elements preserved as specified

6. Other Requirements

6.1 Database Requirements

- Local SQLite with WAL mode
- Schema: system*node*config, local*audit*logs, offline*sync*queue
- Indexes on lookup fields
- Foreign key constraints enabled

6.2 Backup and Recovery

- WAL-safe snapshot via flare-backup.sh

- Package DB + keys + config
- Timestamped backups
- Offline operation

6.3 Branding Requirements

- FLARE logos (black/white) theme-aware
- BlueRhea Technologies logo in reserved About slot (SVG)
- Logos distributed to all UI/API/bridge/static paths
- Django Admin branded

6.4 Deployment Requirements

- systemd units for flared and bridge
- Docker compose for local testing
- Local multi-process runner script
- Makefile for common tasks

6.5 Documentation Requirements

- Technical spec (Flare.md)
- README with full setup
- BUILD.md
- Bridge README
- This SRS document

7. Appendices

7.1 Appendix A: IPC Command Set

Command	Params	Response
ping	{}	{pong: timestamp}
status	{}	{engine, socket, storage, branding}
health	{}	{status}
enqueue	{entity_type, payload}	{ok}
pull_queue	{limit}	{items[]}

7.2 Appendix B: Data Model

See Django models and SQLite DDL in technical specification.

7.3 Appendix C: Service Endpoints

- Frontend: http://127.0.0.1:5173
- Django: http://127.0.0.1:8000
- Bridge: https://127.0.0.1:8443
- Flared: /tmp/flared.sock

7.4 Appendix D: Natural Commit Format Examples

- CASUAL_SHORT: "update 3 project files and fix edge cases"
- CONVENTIONAL: "refactor(core): optimize working directory assets (3 files)"
- TWO_LINE: "refactor local module interfaces" + blank line + body

7.5 Appendix E: Glossary of Key Terms

Offline-first, local-first, zero-trust, CRDT, node-locking, WAL, IPC.

4.8 System Feature Detail - Component 8

4.8.1 Purpose

This section provides detailed requirements and design considerations for component 8 of the FLARE system architecture.

4.8.2 Functional Requirements

- FR-81: Component 8 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-82: Component 8 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-83: Component 8 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-84: Component 8 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-85: Component 8 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.8.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.8.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 8.

4.8.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.9 System Feature Detail - Component 9

4.9.1 Purpose

This section provides detailed requirements and design considerations for component 9 of the FLARE system architecture.

4.9.2 Functional Requirements

- FR-91: Component 9 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-92: Component 9 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-93: Component 9 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-94: Component 9 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-95: Component 9 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.9.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.9.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 9.

4.9.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.10 System Feature Detail - Component 10

4.10.1 Purpose

This section provides detailed requirements and design considerations for component 10 of the FLARE system architecture.

4.10.2 Functional Requirements

- FR-101: Component 10 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-102: Component 10 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-103: Component 10 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-104: Component 10 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-105: Component 10 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.10.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.10.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 10.

4.10.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.11 System Feature Detail - Component 11

4.11.1 Purpose

This section provides detailed requirements and design considerations for component 11 of the FLARE system architecture.

4.11.2 Functional Requirements

- FR-111: Component 11 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-112: Component 11 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-113: Component 11 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-114: Component 11 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-115: Component 11 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.11.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.11.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 11.

4.11.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.12 System Feature Detail - Component 12

4.12.1 Purpose

This section provides detailed requirements and design considerations for component 12 of the FLARE system architecture.

4.12.2 Functional Requirements

- FR-121: Component 12 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-122: Component 12 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-123: Component 12 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.

- FR-124: Component 12 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-125: Component 12 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.12.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.12.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 12.

4.12.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.13 System Feature Detail - Component 13

4.13.1 Purpose

This section provides detailed requirements and design considerations for component 13 of the FLARE system architecture.

4.13.2 Functional Requirements

- FR-131: Component 13 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-132: Component 13 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-133: Component 13 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-134: Component 13 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-135: Component 13 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.13.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.13.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 13.

4.13.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.14 System Feature Detail - Component 14

4.14.1 Purpose

This section provides detailed requirements and design considerations for component 14 of the FLARE system architecture.

4.14.2 Functional Requirements

- FR-141: Component 14 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-142: Component 14 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-143: Component 14 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-144: Component 14 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-145: Component 14 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.14.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.14.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 14.

4.14.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.15 System Feature Detail - Component 15

4.15.1 Purpose

This section provides detailed requirements and design considerations for component 15 of the FLARE system architecture.

4.15.2 Functional Requirements

- FR-151: Component 15 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-152: Component 15 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-153: Component 15 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-154: Component 15 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-155: Component 15 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.15.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.15.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 15.

4.15.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.16 System Feature Detail - Component 16

4.16.1 Purpose

This section provides detailed requirements and design considerations for component 16 of the FLARE system architecture.

4.16.2 Functional Requirements

- FR-161: Component 16 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-162: Component 16 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-163: Component 16 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-164: Component 16 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-165: Component 16 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.16.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.16.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 16.

4.16.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.17 System Feature Detail - Component 17

4.17.1 Purpose

This section provides detailed requirements and design considerations for component 17 of the FLARE system architecture.

4.17.2 Functional Requirements

- FR-171: Component 17 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-172: Component 17 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-173: Component 17 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.

- FR-174: Component 17 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-175: Component 17 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.17.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.17.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 17.

4.17.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.18 System Feature Detail - Component 18

4.18.1 Purpose

This section provides detailed requirements and design considerations for component 18 of the FLARE system architecture.

4.18.2 Functional Requirements

- FR-181: Component 18 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-182: Component 18 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-183: Component 18 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-184: Component 18 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-185: Component 18 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.18.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.18.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 18.

4.18.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.19 System Feature Detail - Component 19

4.19.1 Purpose

This section provides detailed requirements and design considerations for component 19 of the FLARE system architecture.

4.19.2 Functional Requirements

- FR-191: Component 19 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-192: Component 19 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-193: Component 19 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-194: Component 19 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-195: Component 19 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.19.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.19.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 19.

4.19.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.20 System Feature Detail - Component 20

4.20.1 Purpose

This section provides detailed requirements and design considerations for component 20 of the FLARE system architecture.

4.20.2 Functional Requirements

- FR-201: Component 20 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-202: Component 20 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-203: Component 20 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-204: Component 20 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-205: Component 20 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.20.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.20.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 20.

4.20.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.21 System Feature Detail - Component 21

4.21.1 Purpose

This section provides detailed requirements and design considerations for component 21 of the FLARE system architecture.

4.21.2 Functional Requirements

- FR-211: Component 21 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-212: Component 21 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-213: Component 21 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-214: Component 21 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-215: Component 21 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.21.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.21.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 21.

4.21.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.22 System Feature Detail - Component 22

4.22.1 Purpose

This section provides detailed requirements and design considerations for component 22 of the FLARE system architecture.

4.22.2 Functional Requirements

- FR-221: Component 22 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-222: Component 22 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-223: Component 22 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.

- FR-224: Component 22 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-225: Component 22 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.22.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.22.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 22.

4.22.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.23 System Feature Detail - Component 23

4.23.1 Purpose

This section provides detailed requirements and design considerations for component 23 of the FLARE system architecture.

4.23.2 Functional Requirements

- FR-231: Component 23 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-232: Component 23 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-233: Component 23 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-234: Component 23 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-235: Component 23 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.23.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.23.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 23.

4.23.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.24 System Feature Detail - Component 24

4.24.1 Purpose

This section provides detailed requirements and design considerations for component 24 of the FLARE system architecture.

4.24.2 Functional Requirements

- FR-241: Component 24 shall implement requirement specification item 1 with defined behavior, inputs, outputs, and error handling.
- FR-242: Component 24 shall implement requirement specification item 2 with defined behavior, inputs, outputs, and error handling.
- FR-243: Component 24 shall implement requirement specification item 3 with defined behavior, inputs, outputs, and error handling.
- FR-244: Component 24 shall implement requirement specification item 4 with defined behavior, inputs, outputs, and error handling.
- FR-245: Component 24 shall implement requirement specification item 5 with defined behavior, inputs, outputs, and error handling.

4.24.3 Non-Functional Aspects

Performance, security, reliability, and maintainability considerations specific to this component are documented herein.

4.24.4 Interfaces

Internal and external interfaces, data contracts, message formats, and protocol details for component 24.

4.24.5 Validation Criteria

Test cases, acceptance criteria, and verification methods for ensuring correct implementation.

4.25 Advanced Feature Specification - Module 25

4.25.1 Overview

Comprehensive specification for FLARE Module 25 covering architectural design, behavioral contracts, and implementation requirements.

4.25.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.25.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.25.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.25.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.25.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.25.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26 Advanced Feature Specification - Module 26

4.26.1 Overview

Comprehensive specification for FLARE Module 26 covering architectural design, behavioral contracts, and implementation requirements.

4.26.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.26.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27 Advanced Feature Specification - Module 27

4.27.1 Overview

Comprehensive specification for FLARE Module 27 covering architectural design, behavioral contracts, and implementation requirements.

4.27.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.27.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28 Advanced Feature Specification - Module 28

4.28.1 Overview

Comprehensive specification for FLARE Module 28 covering architectural design, behavioral contracts, and implementation requirements.

4.28.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.28.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29 Advanced Feature Specification - Module 29

4.29.1 Overview

Comprehensive specification for FLARE Module 29 covering architectural design, behavioral contracts, and implementation requirements.

4.29.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.29.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30 Advanced Feature Specification - Module 30

4.30.1 Overview

Comprehensive specification for FLARE Module 30 covering architectural design, behavioral contracts, and implementation requirements.

4.30.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.30.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31 Advanced Feature Specification - Module 31

4.31.1 Overview

Comprehensive specification for FLARE Module 31 covering architectural design, behavioral contracts, and implementation requirements.

4.31.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.31.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32 Advanced Feature Specification - Module 32

4.32.1 Overview

Comprehensive specification for FLARE Module 32 covering architectural design, behavioral contracts, and implementation requirements.

4.32.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.32.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33 Advanced Feature Specification - Module 33

4.33.1 Overview

Comprehensive specification for FLARE Module 33 covering architectural design, behavioral contracts, and implementation requirements.

4.33.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.33.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34 Advanced Feature Specification - Module 34

4.34.1 Overview

Comprehensive specification for FLARE Module 34 covering architectural design, behavioral contracts, and implementation requirements.

4.34.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.34.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35 Advanced Feature Specification - Module 35

4.35.1 Overview

Comprehensive specification for FLARE Module 35 covering architectural design, behavioral contracts, and implementation requirements.

4.35.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.35.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36 Advanced Feature Specification - Module 36

4.36.1 Overview

Comprehensive specification for FLARE Module 36 covering architectural design, behavioral contracts, and implementation requirements.

4.36.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.36.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37 Advanced Feature Specification - Module 37

4.37.1 Overview

Comprehensive specification for FLARE Module 37 covering architectural design, behavioral contracts, and implementation requirements.

4.37.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.37.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38 Advanced Feature Specification - Module 38

4.38.1 Overview

Comprehensive specification for FLARE Module 38 covering architectural design, behavioral contracts, and implementation requirements.

4.38.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.38.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39 Advanced Feature Specification - Module 39

4.39.1 Overview

Comprehensive specification for FLARE Module 39 covering architectural design, behavioral contracts, and implementation requirements.

4.39.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.39.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40 Advanced Feature Specification - Module 40

4.40.1 Overview

Comprehensive specification for FLARE Module 40 covering architectural design, behavioral contracts, and implementation requirements.

4.40.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.40.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41 Advanced Feature Specification - Module 41

4.41.1 Overview

Comprehensive specification for FLARE Module 41 covering architectural design, behavioral contracts, and implementation requirements.

4.41.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.41.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42 Advanced Feature Specification - Module 42

4.42.1 Overview

Comprehensive specification for FLARE Module 42 covering architectural design, behavioral contracts, and implementation requirements.

4.42.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.42.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43 Advanced Feature Specification - Module 43

4.43.1 Overview

Comprehensive specification for FLARE Module 43 covering architectural design, behavioral contracts, and implementation requirements.

4.43.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.43.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44 Advanced Feature Specification - Module 44

4.44.1 Overview

Comprehensive specification for FLARE Module 44 covering architectural design, behavioral contracts, and implementation requirements.

4.44.1 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 1. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44.2 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 2. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44.3 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 3. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44.4 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 4. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44.5 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 5. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

4.44.6 Detailed Requirements

Detailed functional, operational, and interface requirements for submodule 6. This includes input validation, processing logic, output formatting, exception handling, state management, concurrency control, and data integrity constraints.

8. System Architecture Details

8.1 Architectural View 1

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 1. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.2 Architectural View 2

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 2. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.3 Architectural View 3

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 3. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.4 Architectural View 4

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 4. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.5 Architectural View 5

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 5. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.6 Architectural View 6

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 6. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.7 Architectural View 7

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 7. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.8 Architectural View 8

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 8. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.9 Architectural View 9

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 9. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.10 Architectural View 10

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 10. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.11 Architectural View 11

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 11. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.12 Architectural View 12

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 12. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.13 Architectural View 13

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 13. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.14 Architectural View 14

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 14. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

8.15 Architectural View 15

Component interactions, data flows, deployment topology, scaling considerations, resilience patterns, and operational boundaries for FLARE subsystem 15. Detailed diagrams described in textual form with interfaces, contracts, and state transitions.

9. Data Design

9.1 Data Entity Specification 1

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.2 Data Entity Specification 2

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.3 Data Entity Specification 3

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.4 Data Entity Specification 4

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.5 Data Entity Specification 5

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.6 Data Entity Specification 6

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.7 Data Entity Specification 7

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

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Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.9 Data Entity Specification 9

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.10 Data Entity Specification 10

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

9.11 Data Entity Specification 11

Entity definitions, attributes, relationships, constraints, validation rules, storage formats, indexing strategies, retention policies, and archival considerations.

10. Test & Verification

10.1 Test Case Category 1

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.2 Test Case Category 2

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.3 Test Case Category 3

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.4 Test Case Category 4

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.5 Test Case Category 5

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.6 Test Case Category 6

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.7 Test Case Category 7

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.8 Test Case Category 8

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.9 Test Case Category 9

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.10 Test Case Category 10

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

10.11 Test Case Category 11

Unit, integration, system, acceptance, regression, performance, security, and edge-case testing with entry/exit criteria and traceability.

11. Security & Compliance

11.1 Security Control 1

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.2 Security Control 2

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.3 Security Control 3

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.4 Security Control 4

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.5 Security Control 5

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.6 Security Control 6

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Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.10 Security Control 10

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

11.11 Security Control 11

Threat model considerations, control objectives, implementation details, verification methods, and compliance mapping.

12. Detailed Implementation Requirements

12.1 Requirement Set 1

This section details requirement set 1 including scope, rationale, inputs, processing, outputs, constraints, dependencies, risks, and mitigation strategies for FLARE system implementation. Complete specification with traceability to system features and non-functional requirements.

12.2 Requirement Set 2

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16. Detailed Implementation Requirements

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